

Just saw a tiktok claiming AI isn't depleting water "because of the water cycle"

Posted by u/Disastrous-Code-9699 • 0 points • 13 comments

Jesus fuckin H Christ we are so screwed if this is the average person's intelligence

Comments

u/Sindaj • 1 points • 2026-01-14 21:37 UTC

Microparticles don't get taken out of the water through the water cycle.

That is why acid rain became aa thing.

Why isn't acid rain a thing anymore? Because we implemented Environmental Protections and water filtration systems to take Microparticles out of the water before it goes back into the water cycle.

And the current US administration is doing away with alot of those Environmental Protections, and the datacenters are being built without those systems in place.

So not looking forward to a future with water contaminated with microparticles 🤢

u/Selfish_and_Misled • 1 points • 2026-01-14 21:13 UTC

Water is infinite, just like success. Also I am dim.

u/Ranger_Aggressive • 3 points • 2026-01-14 19:13 UTC

I once made the joke about global warming melting the ice so and us using that water for ai, that joke still makes more sence then what these AI bros try to push nowadays

u/Important_You_7309 • 3 points • 2026-01-14 18:32 UTC

They keep missing the point that it's not about water being "consumed", it's about potable water being redirected away from communities that need it. Governments worldwide are allowing datacentres to be built in water-scarce communities and the result is a drastic drop in local water pressure for the people who already live there, so drastic that their taps no longer work, the best some can manage is a paltry drip.

Is it the fault of the technology itself? No, but the technology is woefully inefficient, underregulated, and is being championed by literal sociopathic oligarchs who would throw their own siblings into a woodchipper if it yielded a 2% raise in stock price.

u/CoolTransDude1078 • 9 points • 2026-01-14 16:35 UTC

Do they like. Not know that the water cycle needs water?? Water gets evaporated, becomes precipitation, fills lakes, dams, etc, and the cycle starts again. That can't be done without water to evaporate. How can they use that as an argument when they clearly don't understand what they're even saying?! I hope the comments were better at least.

u/Disastrous-Code-9699 • 2 points • 2026-01-15 10:07 UTC

Unfortunately the comments were not and were in support of the OP, at least from what I could tell

u/Moliri-Eremitis • 3 points • 2026-01-14 17:49 UTC

But... the way AI uses water is literally evaporation? It's sprayed over cooling towers to carry heat away in much the same way sweat works. It's not like the water is chemically incorporated into a solid or something.

That said, freshwater is a valuable resource, and you absolutely can evaporate it more quickly than can be replenished, and that's bad. Just because water evaporates and condenses naturally doesn't mean that causing more evaporation is

inherently harmless.

This isn't a question of if using water at all is bad, we use water for all kinds of stuff, it's a question of if the local freshwater supply is capable of supplying sufficient water without becoming stressed in ways that negatively impacts other uses and the environment.

That's certainly not impossible to do with AI. The amount of water that AI uses is not unthinkable massive. It's far less than we spend on agriculture, just for the sake of comparison. Microsoft [made a public commitment] (<https://www.tomshardware.com/tech-industry/artificial-intelligence/microsoft-to-overhaul-ai-data-center-building-with-community-first-approach-says-it-will-be-a-good-neighbor-to-communities-cover-energy-cost-increases-and-replenish-water>) to account for these sort of local impacts just yesterday.

It's worth maintaining some pressure on this issue to ensure that AI doesn't cause needless environmental damage, but it also not an intractable problem inherent to AI that cannot be solved, or will take some kind of scientific breakthrough to address.

u/BagOld5057 • 3 points • 2026-01-14 18:21 UTC

That also doesn't take into account that the water that is evaporated from an area for the most part will not necessarily condense and return to that area. That's how these natural water sources are getting depleted, they are being pumped away and evaporated far faster than they naturally would, and then due to weather patterns that water is being removed from the area before it can rain and return to the source. These data centers are completely exhausting fresh water bodies in ways that do not replenish them.

As for agriculture, I am so unbelievably sick of that substanceless "counter argument." The water used for agriculture produces something of use, and since it is spent primarily in ways that send it directly into the soil (irrigation or livestock urine), that water is largely returned to the groundwater in the area it came from. Generative AI produces nothing useful or necessary, does not operate in a way to sustainably return the water to the region's groundwater, and that's without taking into account the other vast damages involved in its use.

u/Moliri-Eremitis • 0 points • 2026-01-14 19:07 UTC

> That also doesn't take into account that the water that is evaporated from an area for the most part will not necessarily condense and return to that area. That's how these natural water sources are getting depleted, they are being pumped away and evaporated far faster than they naturally would, and then due to weather patterns that water is being removed from the area before it can rain and return to the source.

I agree, and I did directly address that point:

> That said, freshwater is a valuable resource, and you absolutely can evaporate it more quickly than can be locally replenished, and that's bad. Just because water evaporates and condenses naturally doesn't mean that causing more evaporation is inherently harmless.

> As for agriculture, I am so unbelievably sick of that substanceless "counter argument." The water used for agriculture produces something of use, and since it is spent primarily in ways that send it directly into the soil (irrigation or livestock urine), that water is largely returned to the groundwater in the area it came from.

I'll admit to not being an expert on the amount of water that returns to groundwater as part of irrigation, but I do know that a huge percentage of water that a plant takes up is evaporated. It's over 99% according to [this](https://coagmet.colostate.edu/extended_etr_about.php).

That page also states:

> The overall amount of ET (evapotranspiration) for a crop or landscape over an entire growing season is about the same as the seasonal water requirement.

This seems to suggest that the vast majority is evaporated, not returned to groundwater.

Also, at least in the US, a lot of corn is grown for the production of ethanol used in gasoline, not food. This is a double harm, both for its water consumption and because we burn the ethanol and produce greenhouse gasses. Ceasing that through vehicle electrification frees up a _ton_ of water consumption for other things.

> Generative AI produces nothing useful or necessary,

This is inherently subjective. That's fine for arguing that you would personally rather not have freshwater used for AI, but because freshwater is a shared resource it's something that is open for anyone to use how they see fit, as long as it is not taking away from other's ability to use it as well, or harming the environment (another shared resource.)

Maintaining a non-harmful level of extraction is something I agree with, and seems entirely possible due to the levels that are being used, just not in all areas.

There have been cases where the degree of extraction has been disproportionate to the supply and caused problems for others. That's something we should definitely prevent through regulate. It doesn't seem to be a an inherent problem that prevents AI from being used at all, however, just something we need to regulate to ensure it remains proportional and non-harmful.

u/BagOld5057 • 2 points • 2026-01-14 19:37 UTC

Firstly, that's just the water that the plant takes up out of the total amount going back into the soil, and that is the slow rate of natural evaporation I was talking about data centers not accomplishing. That slow evaporation at near-ground level is logically much more easily returned to a more local source than vast quantities of water evaporated at the altitude of a cooling tower very quickly.

Secondly, vehicle electrification is faux-sustainability. The power to charge them is produced by fossil fuels and the massive batteries and electrical components are produced from materials harvested through highly environmentally destructive mining. EVs are an exercise in virtue signaling without looking deeper. (By the way, a lot of the components to make AI centers function get their materials through the same exact methods). Even that issue aside because it is already known that automotives across the board are not a sustainable necessity, it is still an apples-to-oranges comparison. Regardless of the unsustainable nature of automotives in any form, they are still a necessary part of a majority of American's lives, while generative AI has incredibly high resource use and environmental damage while at the same time being entirely useless to the overwhelming majority of people.

Similarly, its really not subjective. Food versus another low effort/low quality image, or another instance of nonconsensual pornography, or a derivative song made by mashing together the finished work of real artists that was used for training data. The comparison is night and day, and none of the products of generative AI have even remotely enough value to justify its damage. You saying "as long as it is not taking away from other's ability to use it as well, or harming the environment (another shared resource.)" is spot on, but that is fundamentally not reality. Just look at Memphis, where residential areas are without water supply and have significantly higher electricity costs as a direct result of the data center that operates nearby. Entire aquifers in New Carlisle, IN are totally depleted because of the fresh water demands of the data center that was erected there. It is all well and good to say that we need to regulate to maintain a "non-harmful level of extraction," but that is simply not how it has functioned, is functioning, or will function for a long way into the future. AI will not operate under conditions where it is not a parasite on the communities it's part of. It doesn't seem possible in any capacity, and is not happening. The energy demands to maintain base level operation, the water demands to keep the servers from overheating, and the space demands chewing up perfectly viable agricultural land do not indicate any ability to coexist with nearby communities.

u/Moliri-Eremitis • 1 points • 2026-01-14 21:43 UTC

\> Firstly, that's just the water that the plant takes up out of the total amount going back into the soil, and that is the slow rate of natural evaporation I was talking about data centers not accomplishing. That slow evaporation at near-ground level is logically much more easily returned to a more local source than vast quantities of water evaporated at the altitude of a cooling tower very quickly.

I think you may be envisioning something like the big, bulbous cooling towers used in nuclear power, which is fair, that's the sort of cooling tower that is in the public consciousness, but that is not the sort of cooling tower used by datacenters. The type of cooling tower used by datacenters is the same sort used in office building HVAC systems and that you see on the tops of buildings. It's not necessarily short, but it's not the sort of massive hunk of concrete you can see for miles like nuclear cooling towers.

Many cooling tower designs also capture, re-condense, and re-use a lot (but not 100%) of the evaporated water. You may have seen videos on TikTok showing the fill packaging inside of one of these cooling towers and the large quantities of water passing over them, along with the implication that all the water shown is immediately lost to the environment. What is frequently not shown is that in many cooling tower designs the liquid water passing over the fill packaging is then immediately re-collected and re-used directly below where the video was being filmed.

A good, concise, simple overview of these types of cooling towers can be seen [\[here\]](https://www.youtube.com/watch?v=UzHJWNL2OtM)(<https://www.youtube.com/watch?v=UzHJWNL2OtM>).

I'm also not sure I agree that evaporation from plants in a local region is likely to return water to that region. The fact that water evaporated from the ocean (inherently at sea level) and travels hundreds of miles inland to condense as rain seems to suggest that it can take extended periods of time and great distances before evaporated water will encounter a high pressure zone and precipitate out.

\> Secondly, vehicle electrification is faux-sustainability. The power to charge them is produced by fossil fuels and the massive batteries and electrical components are produced from materials harvested through highly environmentally destructive mining. EVs are an exercise in virtue signaling without looking deeper.

I would strongly disagree that vehicle electrification is faux sustainability. You do raise a good point that electricity production in many regions is not yet green, which is why vehicle electrification _alone_ would be faux sustainability, but electrifying vehicles and then changing nothing else was never the point.

Vehicle electrification is one part in the overall decarbonization of electricity generation, which is fully possible, in progress, and a necessary part of the green energy transition. Because vehicles remain on the road for decades and required technological advancement to achieve sufficient energy density it made sense to start that process in parallel with decarbonizing the electrical grid itself.

You are, of course, correct that mining is not a zero-impact activity, but the impact of limited, local habitat disruption is a different sort of impact with a different scope and degree of severity than the urgent issue of global climate change caused by greenhouse gas emissions. We _must_ solve the climate issue, and in the very near future.

I have more to say on your remaining points, but I'm at the

character limit and this is already too big of a text wall. I'll come back for the rest.

u/Moliri-Eremitis • 1 points • 2026-01-15 01:02 UTC

Continuation of the post:

\> Even that issue aside because it is already known that automotives across the board are not a sustainable necessity, it is still an apples-to-oranges comparison. Regardless of the unsustainable nature of automotives in any form, they are still a necessary part of a majority of American's lives

I am curious about your sources on the claim of automobiles being unsustainable in any form and not being a sustainable necessity? Even if we completely re-architected cities to be entirely walkable we would still need to move things like produce from the areas where it can be grown to areas where it will be consumed.

Moving goods will require energy no matter what is used to transport the them. It also seems unlikely we could place all of humanity close enough to food production that some form of transportation would be unnecessary.

Even if we did manage to put humanity within walking distance of food production and put everything in a person's life within walking distance it would require massive amounts of construction, which has its own degree of impact. The density requirement alone would necessitate huge amounts of concrete which produces all kinds of emissions.

I'm curious what approach besides vehicle electrification you think is workable to address the transportation sector's contribution to climate change?

\> Similarly, its really not subjective. Food versus another low effort/low quality image (...)

As addressed, not all current agricultural activity is food production. \[40% of all corn grown in the US is used for fuel ethanol\] (<https://www.ers.usda.gov/publications/pub->

details?pubid=105761), and if we do not electrify vehicles I don't see how that is going to be reduced any time soon. We could use more oil, but that increases carbon emissions.

Corn utilized \[almost 83 million acres of land in 2024\] (<https://www.fjdynamics.com/blog/industry-insights-65/crops-in-america-578>), so a little over 33 million acres used simply for fuel production. That's over 51,000 square miles, roughly the size of Alabama, of irrigated corn used for fuel, not food.

It is also not my reading of the previously quoted section that the majority of water used in crop production returns to the ground. Specifically, this:

\> The overall amount of ET (evapotranspiration) for a crop or landscape over an entire growing season is about the same as the seasonal water requirement.

It is my understanding that the "seasonal water requirement" being referenced here is the amount of water that a farmer uses for irrigation. That said, I am open to being wrong on that count, but all the sources I've been able to locate so far don't seem to support the idea that the majority of water used in irrigation returns to groundwater in the local area. Instead, it evaporates and may travel great distances before it precipitates again. If you have a good source that states otherwise I would be happy to be corrected, however.

All of this is to say that my current understanding is:

1. Vehicle electrification is far from faux sustainability, especially if you are concerned about water usage in addition to carbon emissions. Reducing our usage of corn ethanol alone would have a massive impact on water usage.

2. Vehicle electrification is proceeding and will need to continue _along with electrical grid decarbonization_ to avoid unsustainable emissions. As we solve that problem the massive water usage from corn ethanol

production goes away with it, freeing substantial amounts of water for other uses.

I'm open to new information on these subjects, however.

u/Moliri-Eremitis
• 1 points •
2026-01-15 01:33 UTC

Still more continuation:

>
Similarly, its really not subjective. Food versus another low effort/low quality image (...) or another instance of nonconsensual pornography, or a derivative song made by mashing together the finished work of real artists that was used for training data. The comparison is night and day, and none of the

products of generative AI have even remotely enough value to justify its damage.

"Low quality", "derivative", and "not remotely enough value" are all subjective evaluations. You are welcome to that opinion, but quality and degree of value from media is inherently subjective. Personally I greatly enjoy both creating and consuming AI generated images and music, and the fact that there is a lively debate between

the pro and anti AI sides seems to illustrate that there are quite a few people who find value in AI generated output.

> You saying "as long as it is not taking away from other's ability to use it as well, or harming the environment (another shared resource.)" is spot on, but that is fundamentally not reality. Just look at Memphis, where residential areas are without water supply and have significantly higher electricity costs

as a direct result of the data center that operates nearby. Entire aquifers in New Carlisle, IN are totally depleted because of the fresh water demands of the data center that was erected there. It is all well and good to say that we need to regulate to maintain a "non-harmful level of extraction," but that is simply not how it has functioned, is functioning, or will function for a long way into the future.

AI will not operate under conditions where it is not a parasite on the communities it's part of. It doesn't seem possible in any capacity, and is not happening.

I agree that the cases you cited are valid instances where AI has caused negative local impacts to both people and the environment. The existence of cases like these are why I support regulation that prevents these sorts of situations from occurring.

However,

assuming that it _must_ be this way in all instances, now or in then future, is an unsupported assertion in your argument. If we create regulation that ensures protection of the environment and communities and it kills AI in the process, then I would still be in support of the regulation. People and the environment are more important than AI.

I think it's fully possible for AI to operate without thee sorts of impacts. I think it's already happening in

areas besides the negative examples. That said, there are enough variables that it's possible that I'm wrong. Let's regulate AI to ensure that it does not cause these negative impacts and see if it survives. If not, I will celebrate dodging a bullet with you.